

Broadcast Technology UG1

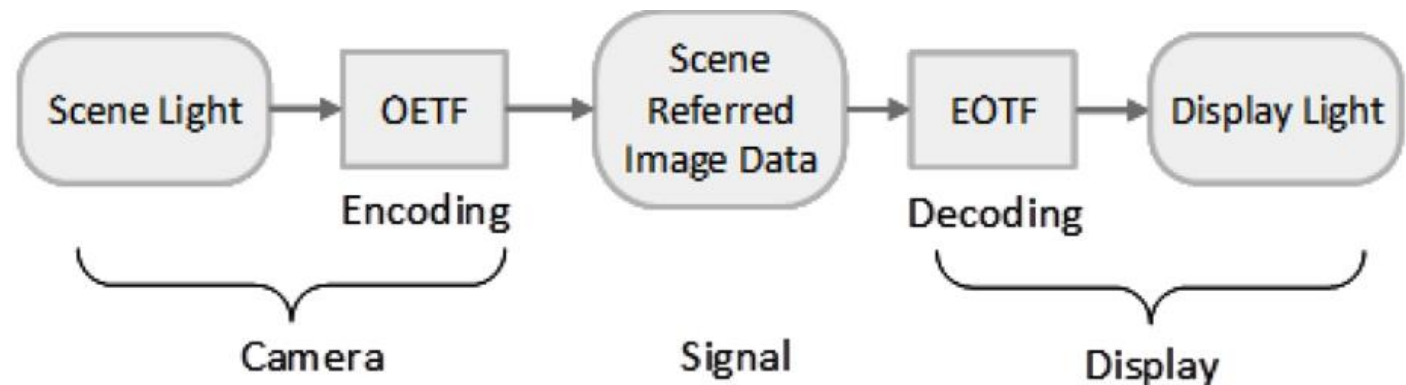
# Gamma and HDR Transfer Functions

# EOTF – Electro-Optical Transfer Function

- In video, capturing, processing, and displaying images involves converting the wide range of light levels from the real world into a limited range of digital values, and then converting those digital values back into light on a screen.
- **Transfer functions** are the mathematical curves or mappings that define these conversions.

## EOTF

- EOTF is the mathematical function that dictates the relationship between the digital signal sent to a display and the resulting light intensity (brightness) emitted by the screen.
- It essentially translates the digital information into the brightness levels we perceive.



# EOTF – Electro–Optical Transfer Function

## EOTF in current standards

- SMPTE 292M, SMPTE 372M, HDMI
  - No EOTF – primarily interface specifications
  - no full details on RGB or YCbCr values should be interpreted in terms of light output
- ITU–R Rec. BT.601, 709, 2020
  - no EOTF – these are encoding specifications
  - these define the parameters like resolution, frame rate, OETF (Opto Electrical Transfer Function)
  - used for image capture

(presentation from Scott Miller,  
Dolby Laboratories)  
<https://www.smpste.org/sites/default/files/2014-05-06-EOTF-Miller-1-2-handout.pdf>



## But image display $\neq$ inverse of image capture

- This is why modern standards like High Dynamic Range (HDR), e.g., ITU–R BT.2100, introduce specific EOTFs.

## How is current EOTF defined?

- Early television and computer monitors used Cathode Ray Tube (CRT) technology. The way a CRT produced light was not directly proportional to the voltage of the incoming signal.
- Human eye perceives light intensity logarithmically.
- **Gamma nonlinearity** refers to the non-linear relationship applied to a digital imaging or video signal to better match human perception of light intensity.
- **Output brightness,  $V_{out}$  = Input voltage,  $V_{in}^\gamma$ .**  $\gamma$  is the exponent that defines the nonlinearity.



Original scene,  
uncorrected



Playback on CRT

Inherent "**gamma**" nonlinearity where darker tones are compressed and brighter tones are expanded.

Finally standardized by ITU-R Rec. BT.1886 (in 2011!!!)

- CRTs almost extinct
- the effort was made to officially document their response curve

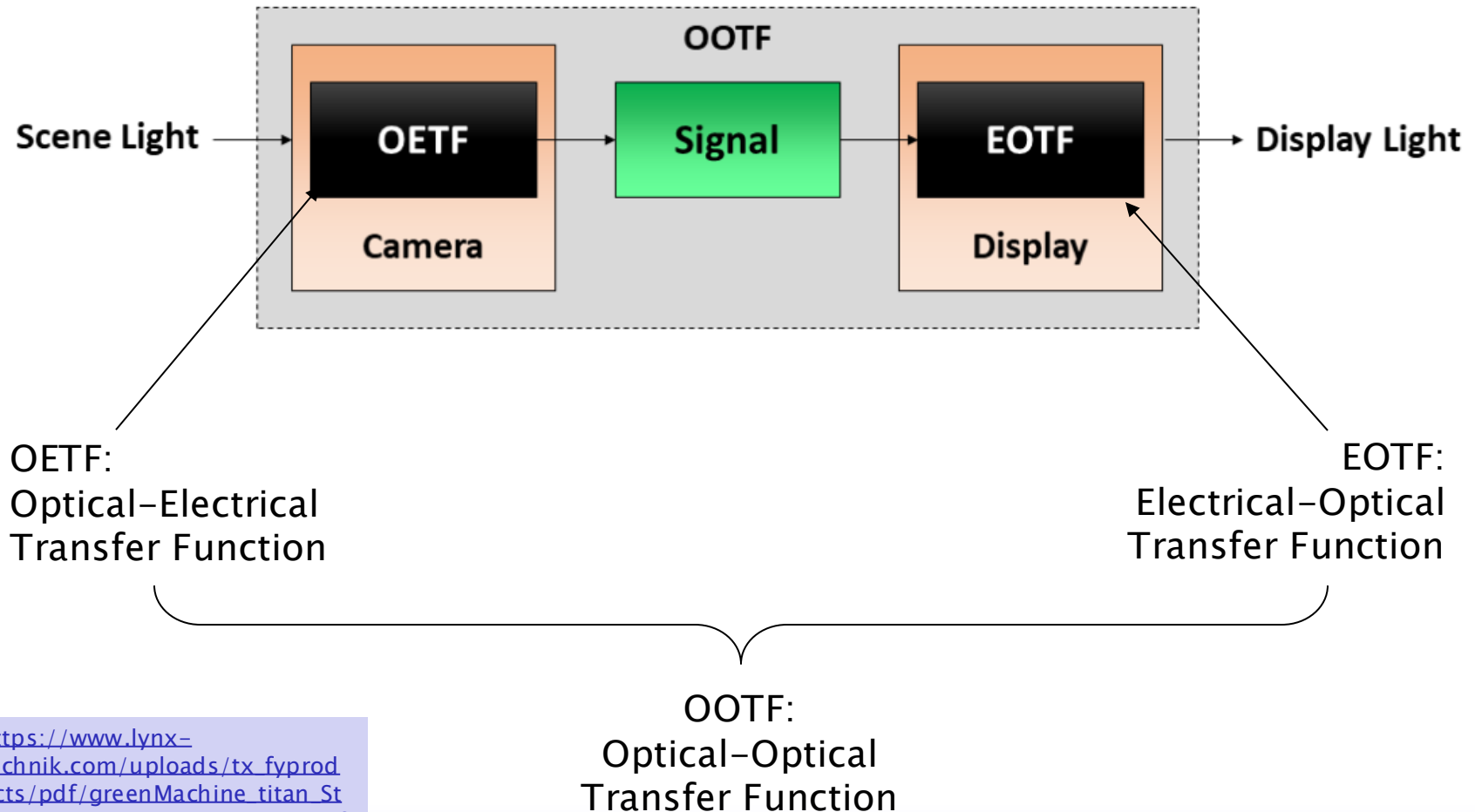
(presentation from Scott Miller,  
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<https://www.smpie.org/sites/default/files/2014-05-06-EOTF-Miller-1-2-handout.pdf>



# The Transfer Function Chain

- A real-world scene is captured, processed as a digital signal, and finally displayed as light on a screen.
- This signal contains the encoded information about the brightness and colour of each pixel in the captured scene.



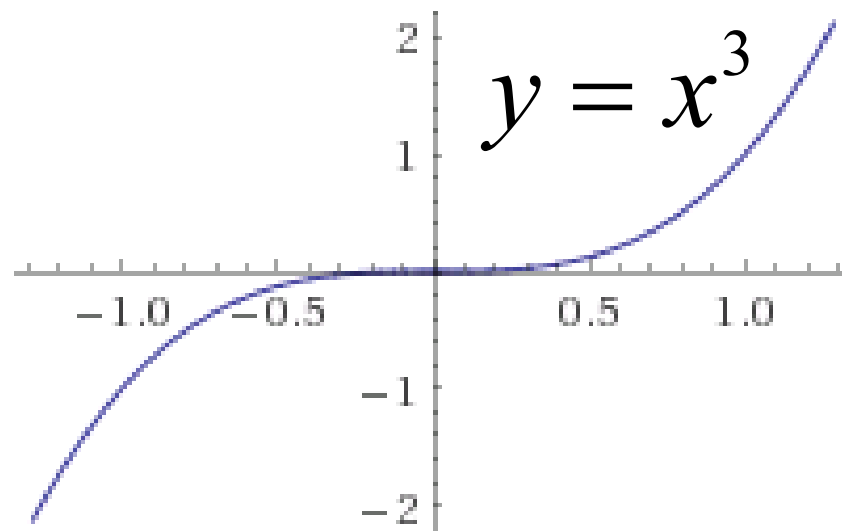
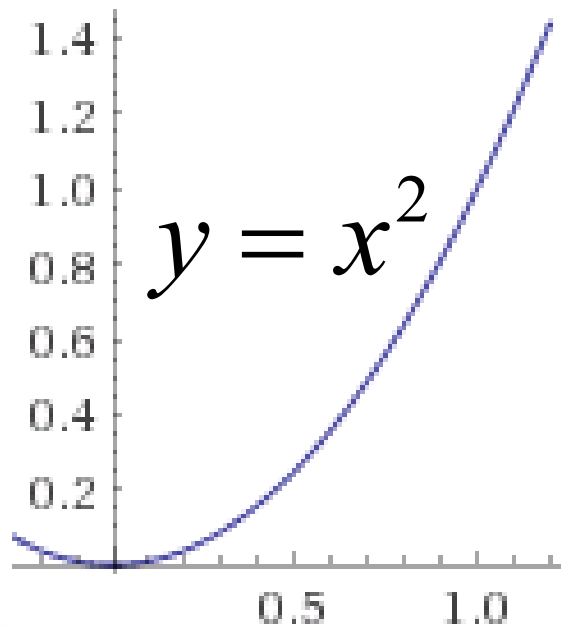
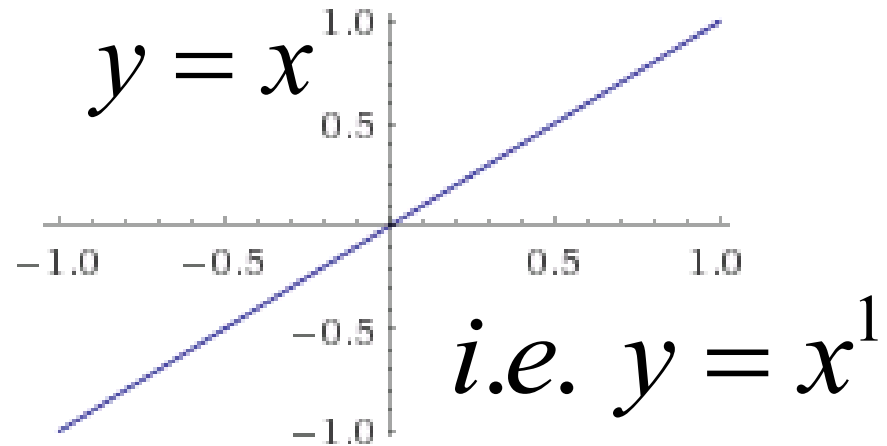
[https://www.lynx-technik.com/uploads/tx\\_fyproductions/pdf/greenMachine\\_titan\\_Static\\_HDR\\_Processing\\_Rev2.0.pdf](https://www.lynx-technik.com/uploads/tx_fyproductions/pdf/greenMachine_titan_Static_HDR_Processing_Rev2.0.pdf)

- **Power Curves**
- **Binary System Capacity**
- **Gamma**

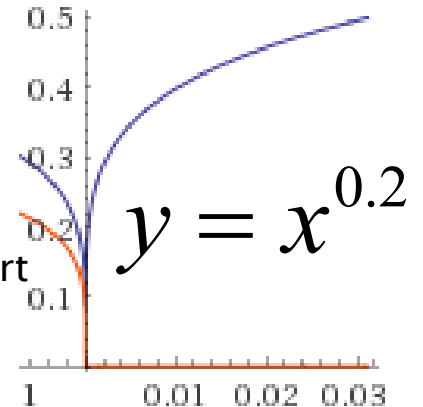
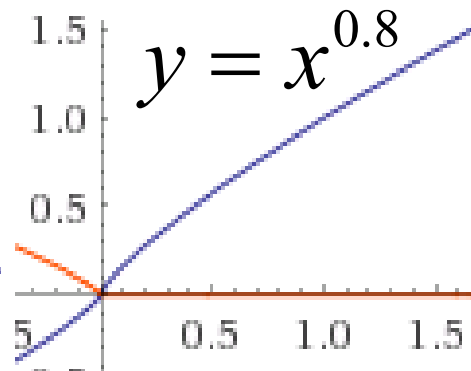
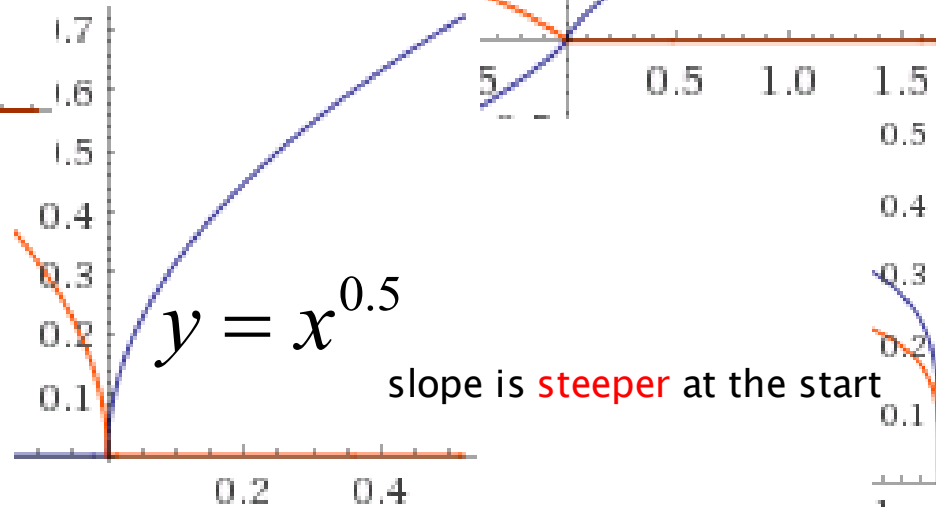
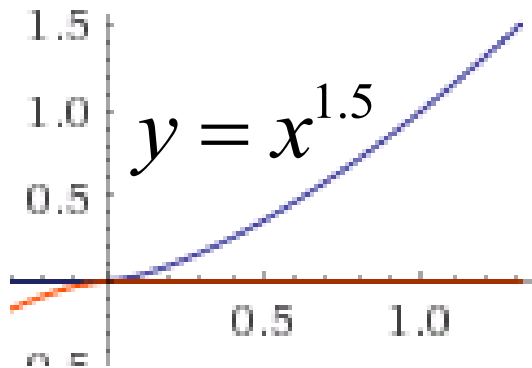
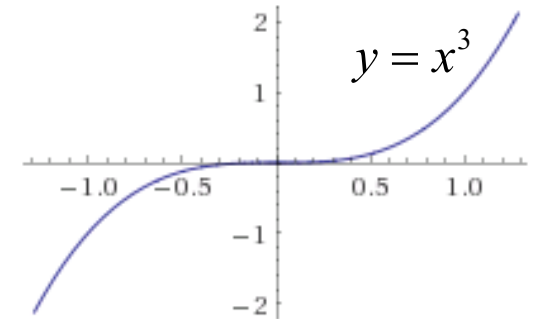
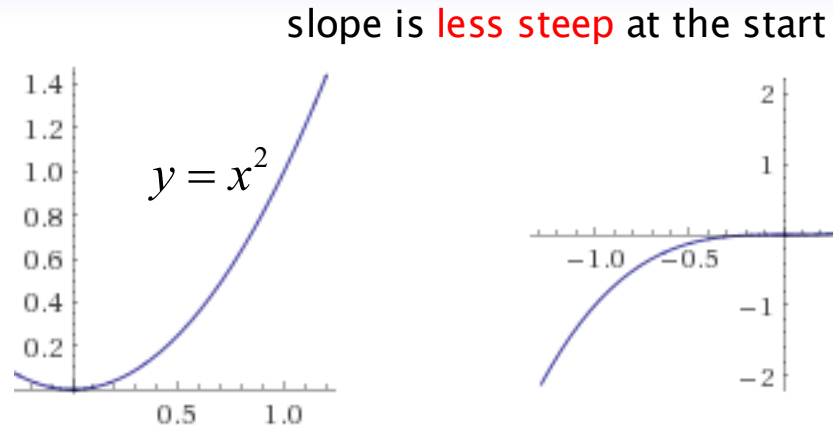
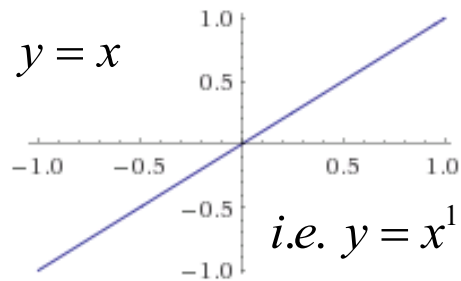
# Power-Based Curves

A power curve is a mathematical function where the output is related to the input raised to a certain power (exponent).

$$V_{out} = V_{in}^{\gamma}$$

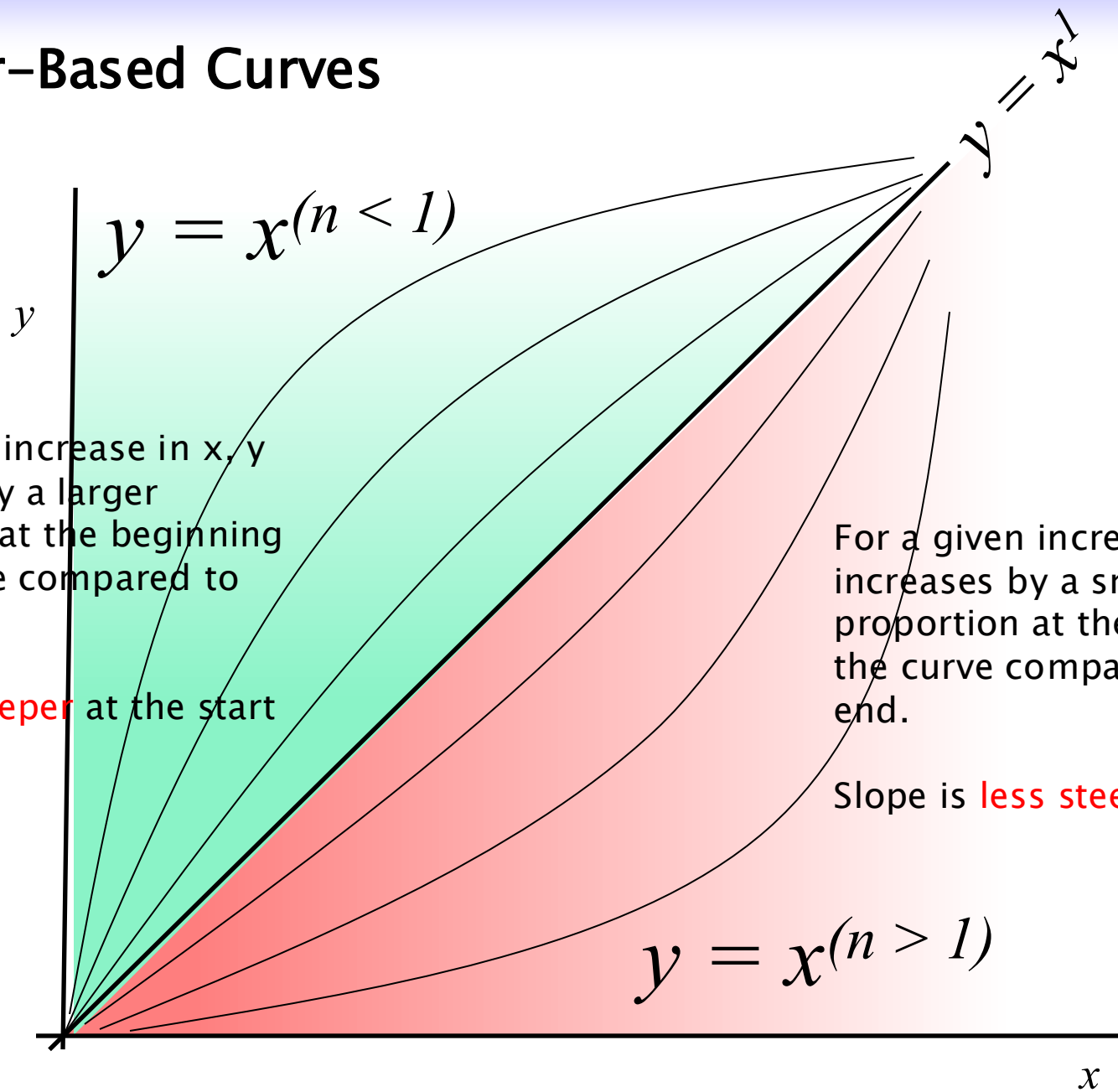


# Power-Based Curves





# Power-Based Curves



# Revision – Counting in Binary

|  |    |        |
|--|----|--------|
|  | 32 | 100000 |
|  | 31 | 11111  |
|  | 30 | 11110  |
|  | 29 | 11101  |
|  | 28 | 11100  |
|  | 27 | 11011  |
|  | 26 | 11010  |
|  | 25 | 11001  |
|  | 24 | 11000  |
|  | 23 | 10111  |
|  | 22 | 10110  |
|  | 21 | 10101  |
|  | 20 | 10100  |
|  | 19 | 10011  |
|  | 18 | 10010  |
|  | 17 | 10001  |
|  | 16 | 10000  |
|  | 15 | 1111   |
|  | 14 | 1110   |
|  | 13 | 1101   |
|  | 12 | 1100   |
|  | 11 | 1011   |
|  | 10 | 1010   |
|  | 9  | 1001   |
|  | 8  | 1000   |
|  | 7  | 111    |
|  | 6  | 110    |
|  | 5  | 101    |
|  | 4  | 100    |
|  | 3  | 11     |
|  | 2  | 10     |
|  | 1  | 1      |
|  | 0  | 0      |

doubling in value  
over previous step

doubling in value  
over previous step

doubling in value  
over previous step

# Exposure / illumination intensity against a binary scale (binary leading zeros filled in)

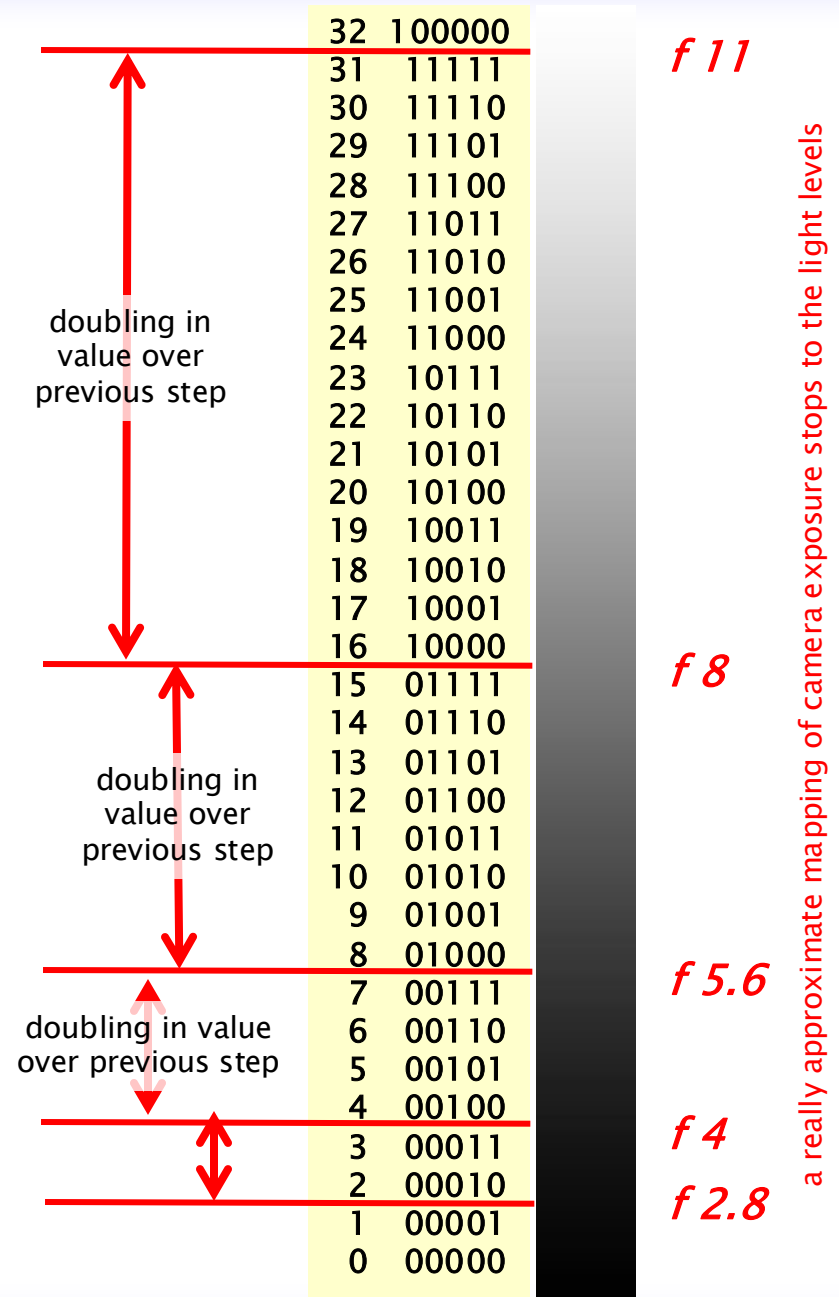
We are mapping real-world light levels onto a digital scale represented by binary numbers.

... we have more binary levels towards the “bright end” (smaller apertures) ...

... but we need more levels at the lower light levels

.. because we perceive illumination differences better at lower light levels

**F-stop** (or f-number) is a measure of the aperture size of a camera lens, which controls the amount of light entering the camera.



# Our Eyes Perceive Brightness Levels In a Logarithmic Manner, not linearly

Imagine one lit candle in a dark room

- If we light fifteen additional candles, one at a time, the sequence of total candles lit looks like this:

• candles – 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16



At sixteenth candle – might assume the room would appear to be sixteen times as bright as it did with one candle lit

- however, **the eye perceives light levels in logarithmic values**
- room appears just 5x as bright compared to the first candle



# Linear and Logarithmic Gamma

## Logarithmic change means

- the apparent change in brightness between one candle and two ... (in our eyes) same as between two candles and four ...
  - four candles to eight ... same brightness increase
  - eight to sixteen ... same brightness increase

• candles – 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16



... the eye  
sees as ...

perceived brightness *increase*  
is the same

A logarithmic scale compresses  
large linear changes into smaller  
perceived changes.

$$1 \text{ to } 2 \text{ candles: } \log_2(2) - \log_2(1) = 1 - 0 = 1$$

$$2 \text{ to } 4 \text{ candles: } \log_2(4) - \log_2(2) = 2 - 1 = 1$$

$$4 \text{ to } 8 \text{ candles: } \log_2(8) - \log_2(4) = 3 - 2 = 1$$

$$8 \text{ to } 16 \text{ candles: } \log_2(16) - \log_2(8) = 4 - 3 = 1$$

# Linear and Logarithmic Gamma

*Looking at it the other way ...*

- In order for the eye to see four equal steps increase of brightness:

- $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$

... we need to light the following numbers of candles:

- $1 \rightarrow 2, 4, 8, 16$



The logarithmic steps get bigger as the room appears brighter (1 to 2 is a difference of 1 candle, but 8 to 16 is a difference of 8 candles)...

- eyes are very good at detecting small steps in dark and mid-tones
- but the brightness steps get much bigger when we look at bright objects

For each step to feel equally brighter:

- 1 candle:  $\log_2(1)=0$
- 2 candles:  $\log_2(2)=1$
- 4 candles:  $\log_2(4)=2$
- 8 candles:  $\log_2(8)=3$
- 16 candles:  $\log_2(16)=4$

*... however, camera sensor has linear response*

- *binary coding effectively* allocates:
  - most (smallest) steps to highlights
  - least (biggest) steps to mid-tones and shadows

# Gamma ( $\gamma$ )

## Aim:

Optimize the Use of Digital Codes in Digital Imaging with Gamma Correction.

The goal is to optimize the allocation of digital codes in a way that better matches human perception, ensuring that the limited range of binary codes (e.g., 0 to 16,384 for a 14-bit sensor or 0 to 65,536 for a 16-bit sensor) is used efficiently to represent light intensity.

## Problem:

- The eye perceives light intensity in a *logarithmic* manner
  - The digital sensor captures light intensity in *linear* manner
- ... and Binary coding stores these linear values directly:

0000000000000000 = 0

0000000000000001 = 1

0000000000000010 = 2

... etc

...

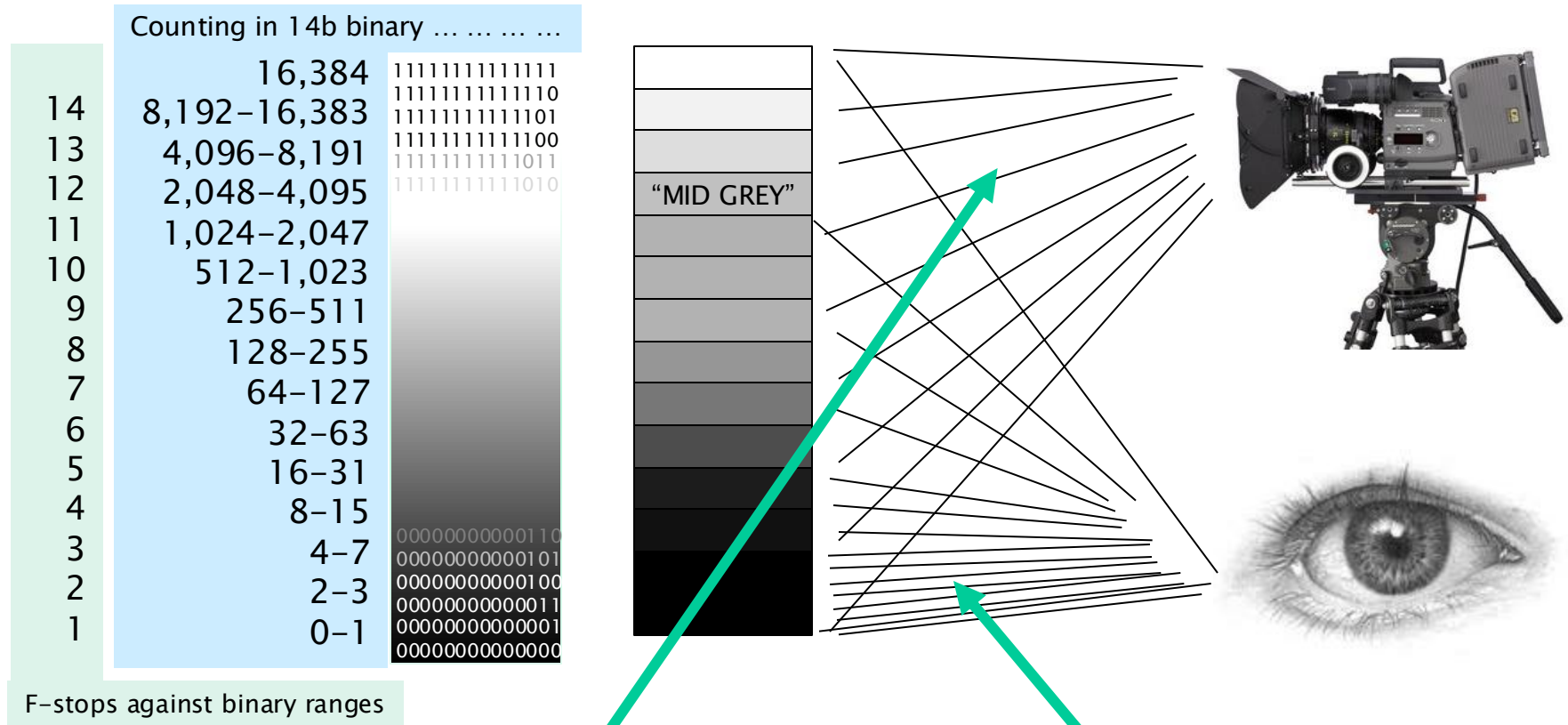
This linear encoding inefficiently allocates codes: It assigns too many codes to highlights (where the eye is less sensitive to changes) and too few to shadows and mid-tones.

to 1111111111111111 = 16,384 ... for a 14 bit sensor

... or 1111111111111111 = 65,536 ... for a 16 bit sensor

To optimize the use of digital codes, we apply gamma correction, which adjusts the linear sensor data to better match the logarithmic nature of human perception.

# Linear Camera / Logarithmic Eye



## Dichotomy

- Camera – linear recording scene illumination
  - applies most coding allowance to the light – tone end of the scale ...
- The Eye – is most effective at distinguishing tonal differences in the darker range of the scale.

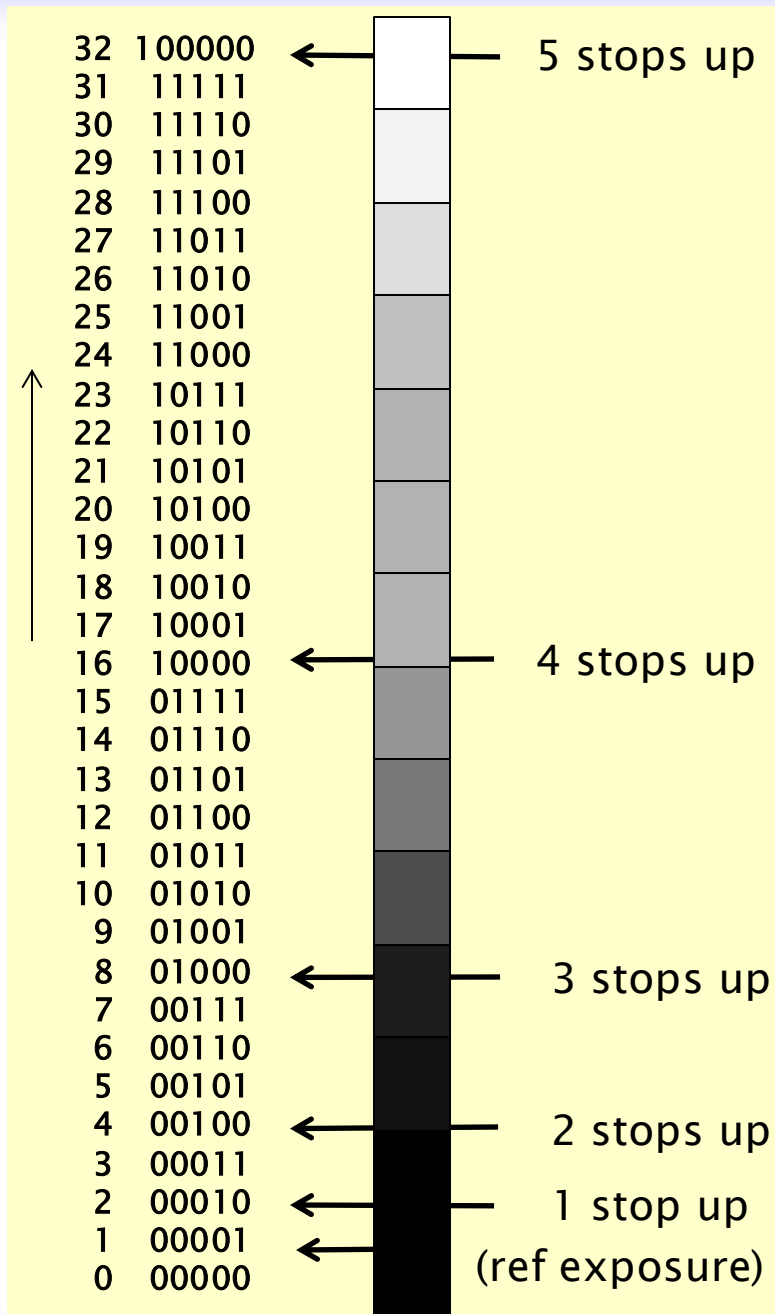


# Camera / Eye Levels

0 – 31 and so on ...  
Linear increase in  
exposure  
.. as seen and recorded  
by sensor



A linear progression of  
exposure levels as the  
sensor captures them.  
The binary numbers  
alongside show a  
potential linear  
encoding of these  
levels.



Exposure stops:  
Linear perception of  
increase in brightness,  
as seen by the eye

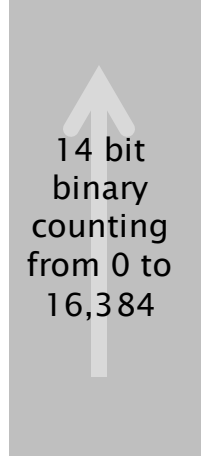


note – for each  
successive stop more  
linear levels are  
recorded by the  
sensor (in fact twice  
as many each stop)  
... but the eye doesn't  
perceive all the  
increasing levels

# Quantization Levels Aligned to Levels of White – Black

For the number of light levels we have: **Each stop represents a halving of the light intensity.**

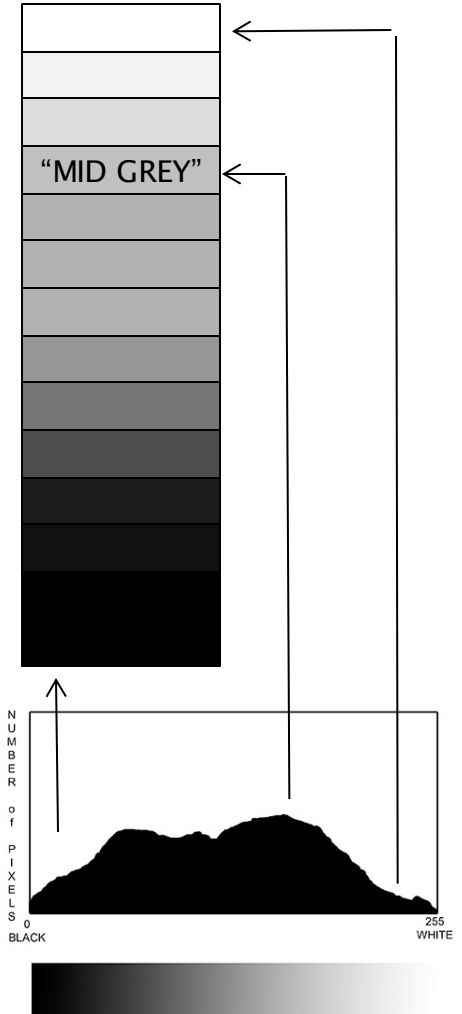
11111111111111  
11111111111110  
11111111111101  
11111111111100



000000000000100  
000000000000011  
000000000000001  
000000000000000

16,384  
8,192–16,383  
4,096–8,191  
2,048–4,095  
1,024–2,047  
512–1,023  
256–511  
128–255  
64–127  
32–63  
16–31  
8–15  
4–7  
2–3  
0–1

totally saturated sensor (maximum white)  
1 stop down from maximum white  
2 stops down from maximum white  
3 stops down from maximum white  
4 stops down from maximum white  
5 stops down from maximum white  
6 stops down from maximum white  
7 stops down from maximum white  
8 stops down from maximum white  
9 stops down from maximum white  
10 stops down from maximum white  
11 stops down from maximum white  
12 stops down from maximum white  
13 stops down from maximum white  
14 stops down from maximum white



- note the large number of bits to store the near-white levels
- low number of bits for darker levels
- note mid-grey at 3 stops down from maximum white
- On a histogram

...upper levels are right-side of the graph ...

**This reinforces why a non-linear transformation like gamma correction is essential.**

# Correction to a Power-Based Curve

Red curve illustrates the Opto-Electrical Transfer Function (OETF) applied during encoding (often in the camera or during post-production).

(Simplistic explanation):  
Input is raised to a power  $1/\gamma$

$$V_{\text{file}} = V_{\text{in}}^{1/\gamma}$$

And output (display must be inverse of this,  $\gamma$ ):

$$V_{\text{display}} = V_{\text{in}}^{\gamma}$$

Actually, the equations may include a system constant  $A$ :

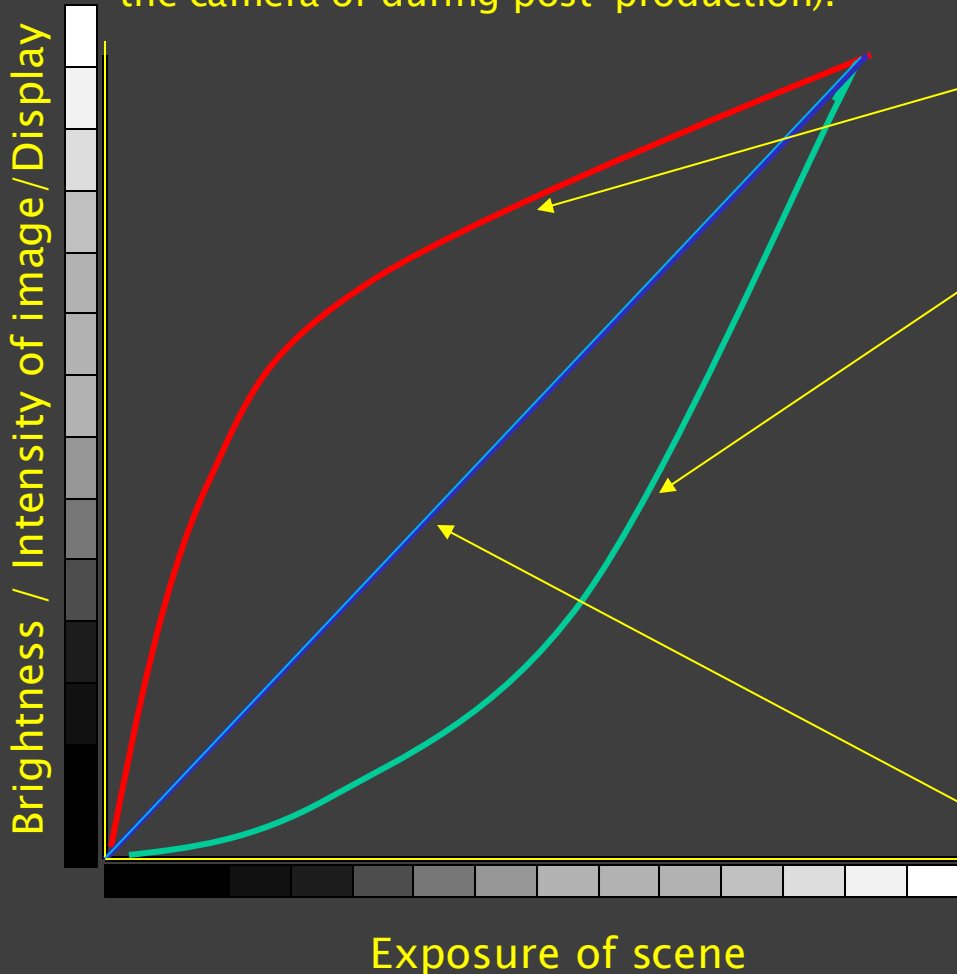
$$V_{\text{file}} = A V_{\text{in}}^{1/\gamma}$$

(where  $A = a \text{ constant}$ )

$$V_{\text{display}} = A V_{\text{in}}^{\gamma}$$

Result:

The eye needs to see the result of the input  $\Rightarrow$  output transfer  
... and should be a straight line  
i.e. the viewer sees the original scene



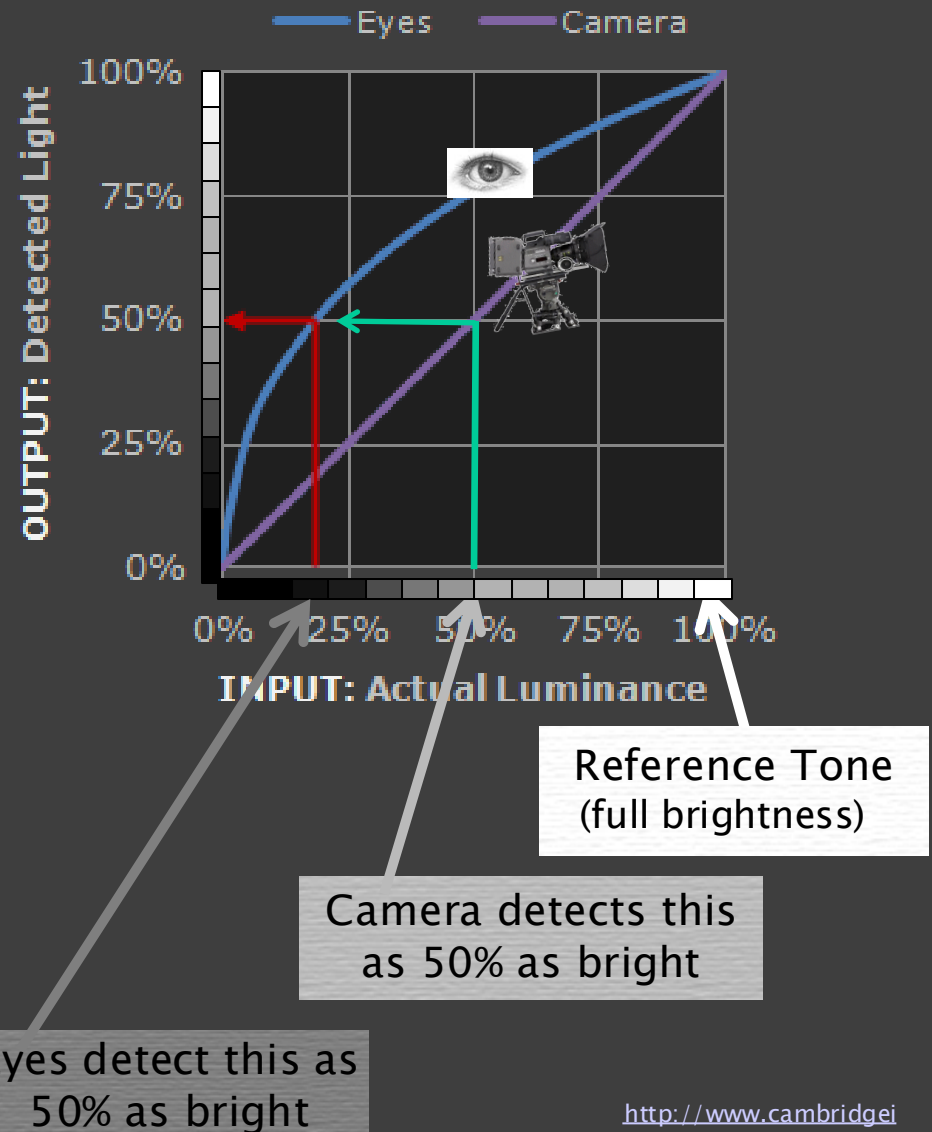
Green curve illustrates the Electro-Optical Transfer Function (EOTF) of the display.

# Eye Response Compared to Camera ... Gamma ( $\gamma$ )

Camera response is linear: a camera sensor outputs a signal directly proportional to the amount of light it receives.

Eyes are much more sensitive to changes in dark tones than we are to *similar changes* in bright tones ... this enables our vision to operate over a broader range of luminance

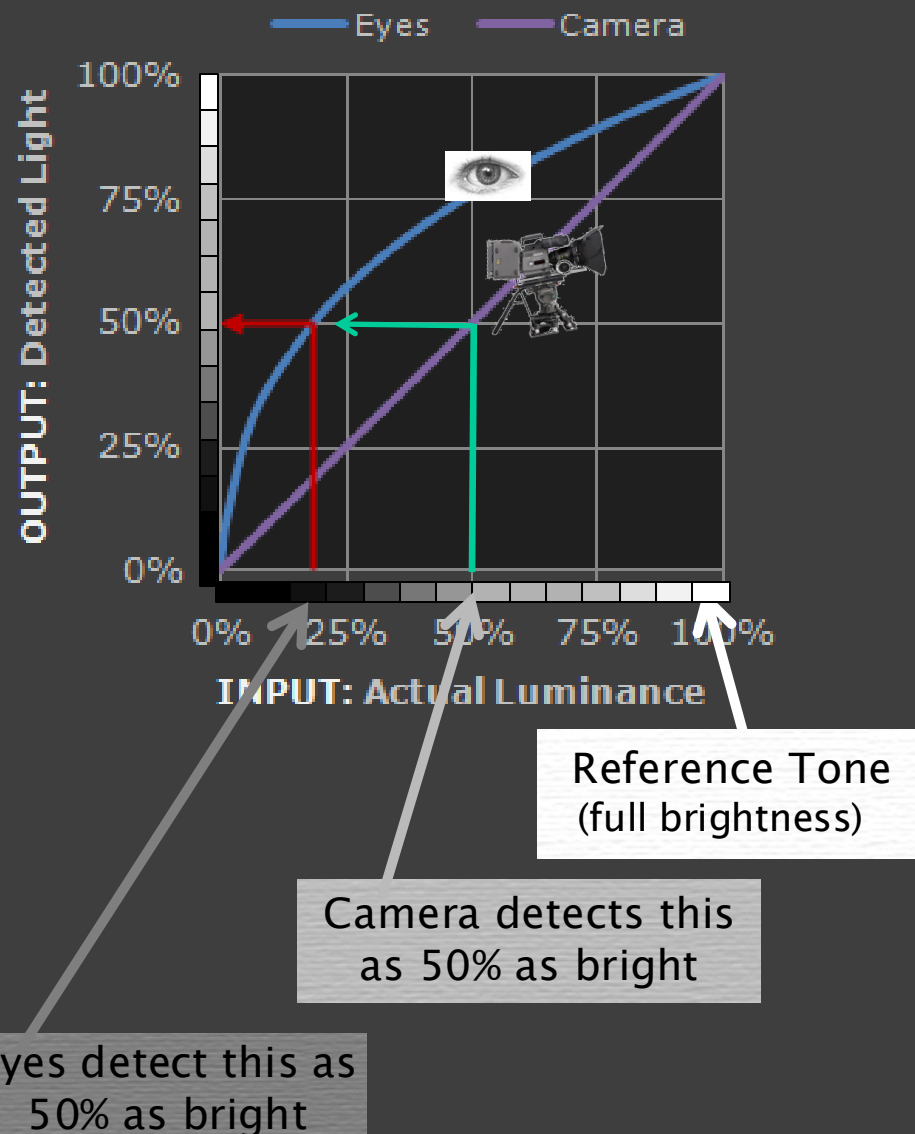
(otherwise – outdoors brightness ranges would be too overwhelming)



# Eye Response Compared to Camera ... Gamma ( $\gamma$ )

Gamma translates between our eye's light sensitivity and that of the camera

- when a digital image is saved, it's gamma encoded
- the file more closely corresponds to what we would perceive in brightness increases



# Gamma ( $\gamma$ ) on a gradient image

Gamma encoded images store tones more efficiently

- redistributes tonal levels closer to how our eyes perceive them



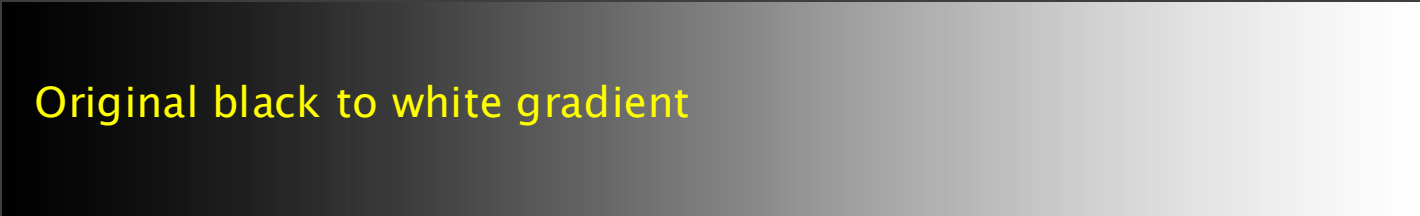
Original black to white gradient

- In a linear system, the change in light intensity is uniform across this gradient.

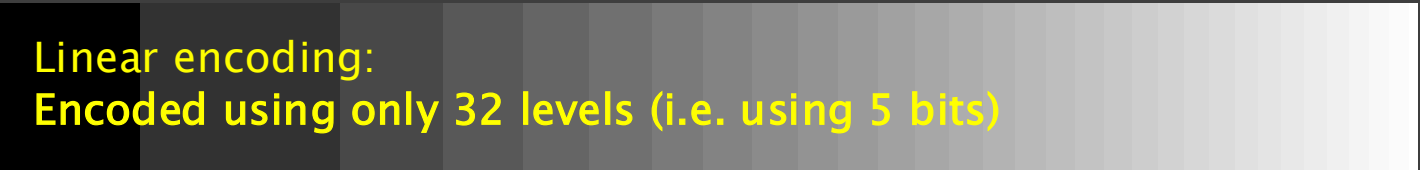
# Gamma ( $\gamma$ ) on a gradient image

Gamma encoded images store tones more efficiently

- redistributes tonal levels closer to how our eyes perceive them



Original black to white gradient



Linear encoding:  
Encoded using only 32 levels (i.e. using 5 bits)

- This smooth gradient is encoded using a limited number of digital levels (only 32, which requires 5 bits of data).
- Because the encoding is linear, these 32 levels are distributed evenly across the entire range from black to white.

# Gamma ( $\gamma$ ) on a gradient image

Gamma encoded images store tones more efficiently

- redistributes tonal levels closer to how our eyes perceive them



Original black to white gradient



Linear encoding:  
**Encoded using only 32 levels (5 bits)**

Gamma encoding: redistributes tones

- This shows the same original gradient encoded with only 32 levels, but this time using gamma correction.
- More bits required to distinguish brighter tones
  - To get smooth gradations in the bright areas with a linear system, you need many more bits (more levels) than are perceptually necessary.
- A shortage of bits for darker tones
  - With a limited number of bits, the darker tones get very few steps, leading to a loss of detail and banding.



# Gamma ( $\gamma$ ) on a gradient image

## Linear encoding

- insufficient levels to describe the dark tones
- excess of levels to describe the bright tones

Original black to white gradient



Linear encoding:  
Encoded using only 32 levels (5 bits)



Gamma encoding: redistributes tones evenly



## Gamma encoded gradient

- distributes tones evenly across the full range  
– *perceptually uniform*

# Gamma ( $\gamma$ ) on a gradient image

Normally for images: use at least 256 levels (8 bits)

Original black to white gradient

... enough to make tones appear smooth and continuous ...

Linear encoding:  
Encoded using only 32 levels (5 bits)

11 bits would be required to avoid image posterization

Gamma encoding: redistributes tones

*11 bits is 8x as many levels as 8 bits*

*8b = 256 levels*

*11b = 2048 levels*

This highlights the massive increase in data required for a linear encoding to match the perceptual smoothness of a gamma-encoded image.

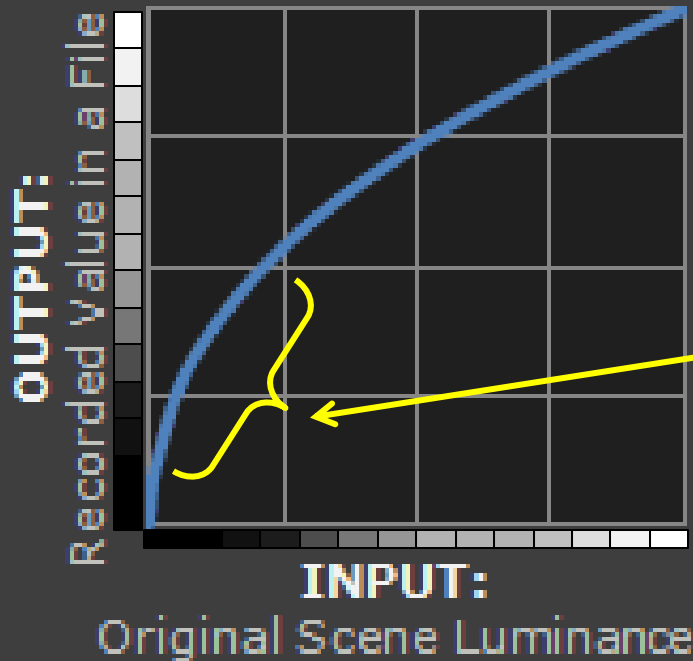
This graded effect is *posterization*

# Gamma Encoding

## - the Image Capture (Recording) Curve

video is recorded with  
Gamma encoding

Linear luminance from the original scene is non-linearly transformed into the recorded values in a digital file.



low-light regions are encoded to expand the range to match the eye response

### Image File Gamma

Depicts an image (e.g. in the RGB colour space) ... which is encoded using a gamma of approx.  $1/2.2 \sim 0.45$

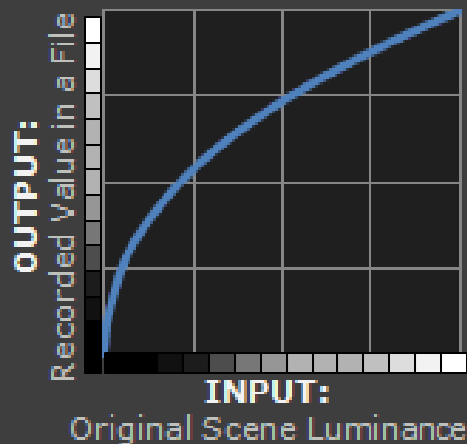
$$V_{\text{file}} = (A)V_{\text{in}}^{1/\gamma}$$

# Gamma Workflow – Encoding and Correction

Gamma encoded image needs "gamma correction" when it is viewed

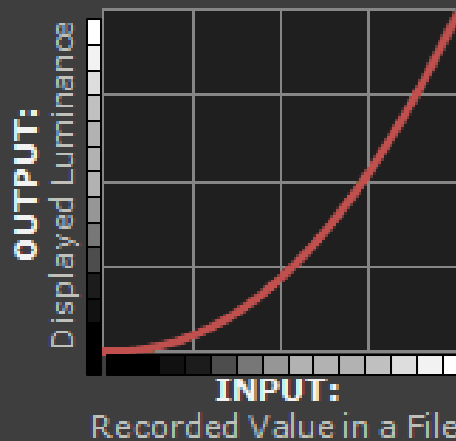
- converts it back into light from the original scene
- gamma encoding is for recording the image – not for displaying the image
- "display gamma" is automatically performed by monitor / video card

video is recorded with  
Gamma encoding



Video File Gamma

Video viewed on  
a monitor

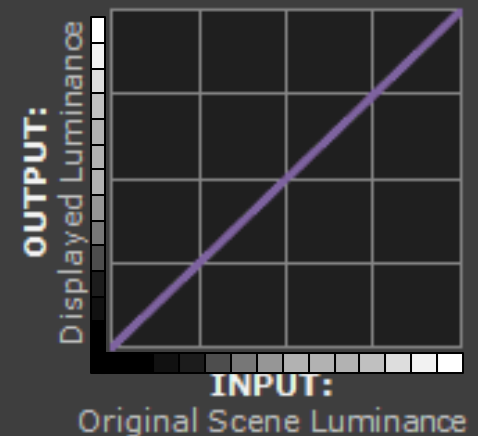


Display Gamma

Depicts a display gamma equal to the standard of 2.2

$$V_{\text{display}} = AV_{\text{in}}^{\gamma}$$

Net Effect



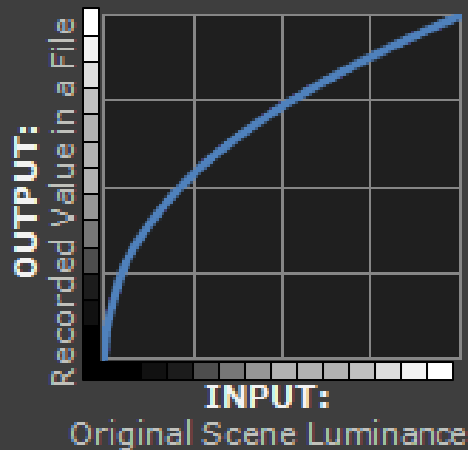
System Gamma  
(Viewing Gamma)

$$\gamma = 1$$

$$\text{i.e. } V_{\text{viewed}} = V_{\text{scene}}$$

# Image Gamma

video is recorded with  
Gamma encoding



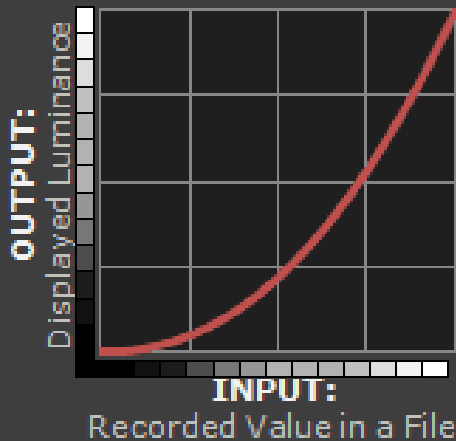
**Video File Gamma**

Is applied:

- in the camera (some cameras apply gamma encoding directly to the captured data before saving it (e.g., when saving JPEGs)), or
- software applied to the RAW image i.e. Gamma encoding is then applied during the RAW processing stage using software when you convert the RAW files into viewable formats like JPEG.

# Display Gamma

Viewed on a  
Computer Monitor



## Display Gamma

Depicts a display  
gamma equal to the  
standard of 2.2

For more modern devices (i.e. not CRT based)

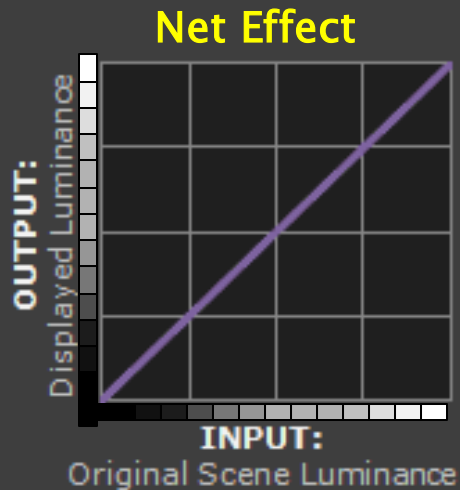
– Could be the net influence of the video  
card and display device

- may be comprised of several gammas
  - (R, G, B)
- ensures that the image isn't unrealistically  
brightened when displayed on screen
- higher display gamma  
gives  
darker image + greater contrast

# System Gamma

The net effect of all gamma values that have been applied to an image

- also referred to as the *viewing gamma*



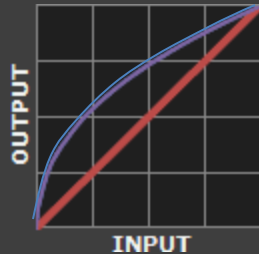
- ideally be close to a straight line (gamma = 1.0)
  - input (the original scene)  
= output (display on the screen)
- A system gamma of 1.0 means that the brightness the viewer sees is directly proportional to the luminance of the original scene.
- However – sometimes set  $> 1$ 
  - A system gamma greater than 1 will increase the contrast in the displayed image, making dark areas darker and bright areas brighter, often leading to a more visually impactful image, even if it's not perfectly accurate
  - ... *but not for a production grade monitor*

**System Gamma**  
(the overall gamma of the end-to-end system)

# Display Gamma

This is altered when performing monitor calibration

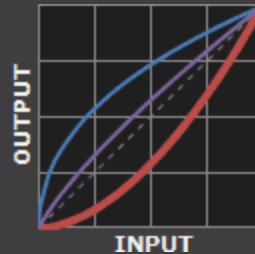
- industry has converged on a standard display gamma of 2.2
- standard gamma encoded image file ———
- display gamma ... .. ———
- overall impact on the image ———
- Incorrect display gamma values lead to images that look either washed out (too low gamma) or overly dark and contrasty (too high gamma).



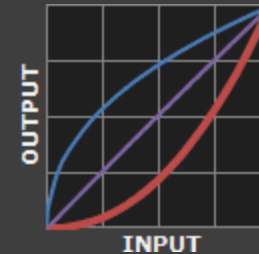
Display Gamma 1.0



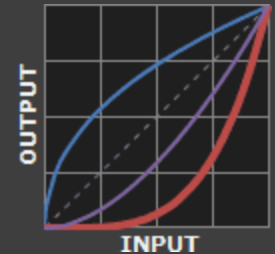
low contrast



1.8



2.2



4.0

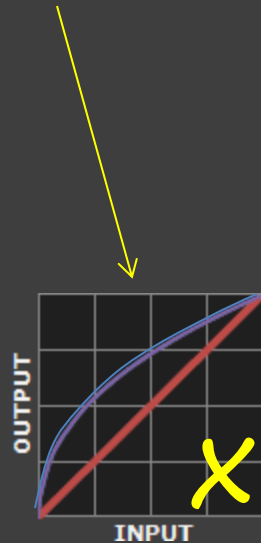


high contrast

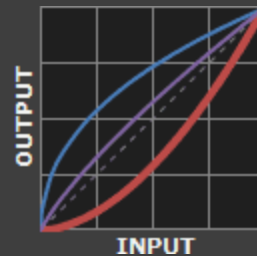


# Display Gamma

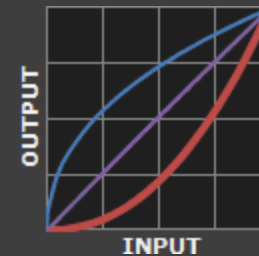
- image brightened substantially because the image gamma is uncorrected by the display gamma
- resulting in an overall system gamma that curves above required straight line of the display



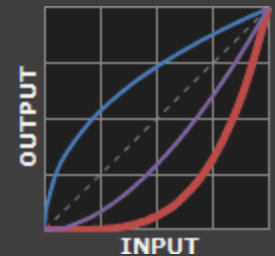
Display Gamma 1.0



1.8



2.2

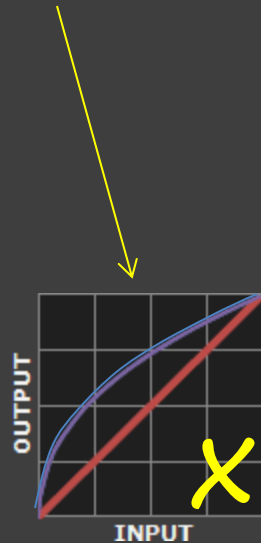


4.0



# Display Gamma

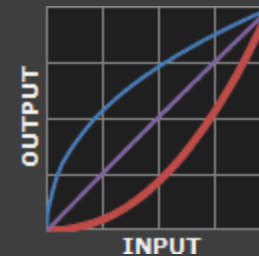
- image brightened substantially because the image gamma is uncorrected by the display gamma
  - resulting in an overall system gamma that curves above required straight line of the display
- display gamma doesn't fully correct for the image file gamma
  - overall system gamma that still curves upward a little (image still a bit too bright)



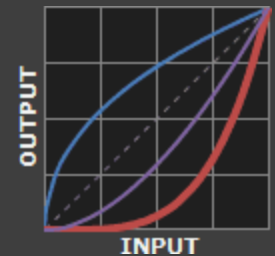
Display Gamma 1.0



1.8



2.2



4.0

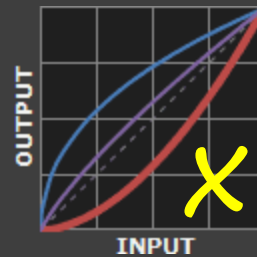


# Display Gamma

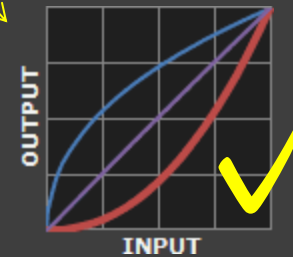
- display gamma exactly corrects the image gamma
- overall linear system gamma



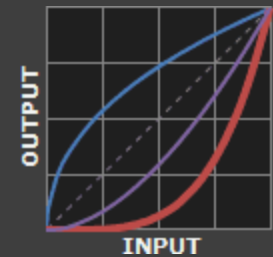
Display Gamma 1.0



1.8



2.2



4.0



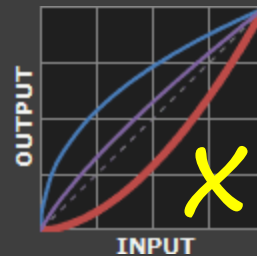
# Display Gamma

- display gamma over-compensates for the image gamma
- overall system gamma that curves downward
- (darkens the image)

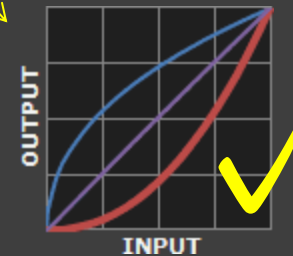
- display gamma exactly corrects the image gamma
- overall linear system gamma



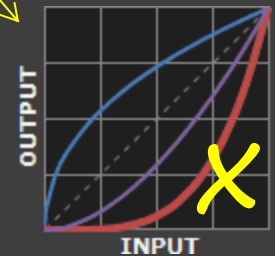
Display Gamma 1.0



1.8



2.2



4.0



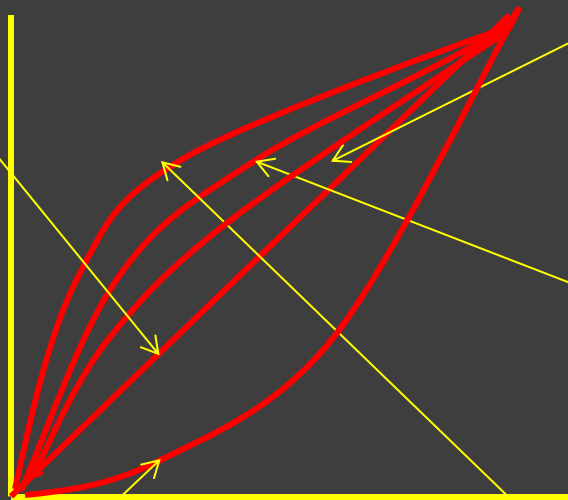
# The Effect Of Gamma Correction On An Image



The original image



$\gamma > 1$   
Shadows darker



$\gamma < 1$   
Shadows lighter



# Display Gamma

*review so far ...*



- Historically, CRT monitors had a native gamma of around 2.5, which coincidentally helped to partially correct for gamma-encoded content without explicit display gamma processing.
- Modern displays (non-CRT) don't have this inherent gamma and rely on processing by the monitor and/or video card to achieve the desired display gamma.
- The industry has converged on a standard display gamma of 2.2 for modern devices.
- Display gamma is usually set by default but can be adjusted during monitor calibration for more accurate viewing.

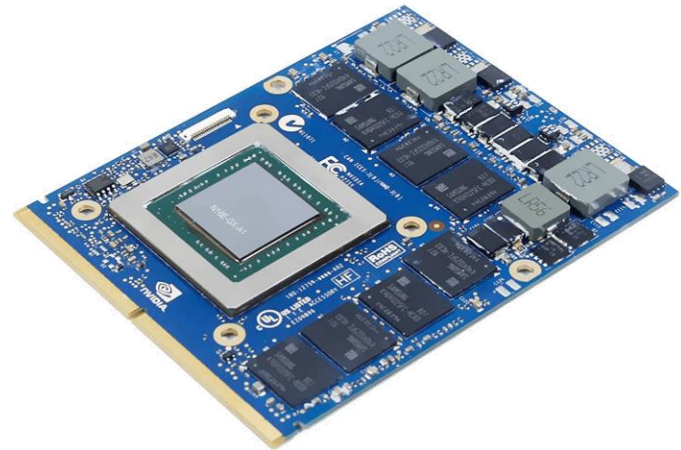
# Display Gamma

Display gamma comprises

the native monitor gamma

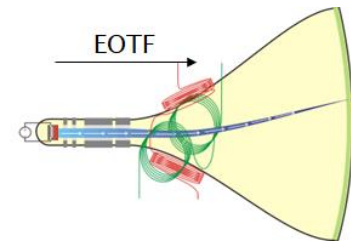
*plus*

any gamma corrections applied within the display itself or by the video card



# HDR (High Dynamic Range) Transfer Functions

- Electro-Optical Transfer Function (EOTF) – the way a display converts digital signals to light – should ideally match how humans perceive brightness.



The current gamma worked well enough for the CRT era

... is a sort of perceptual EOTF

... But only at relatively low luminance levels  
and small contrast ratios (dynamic range)

**But next generation systems**

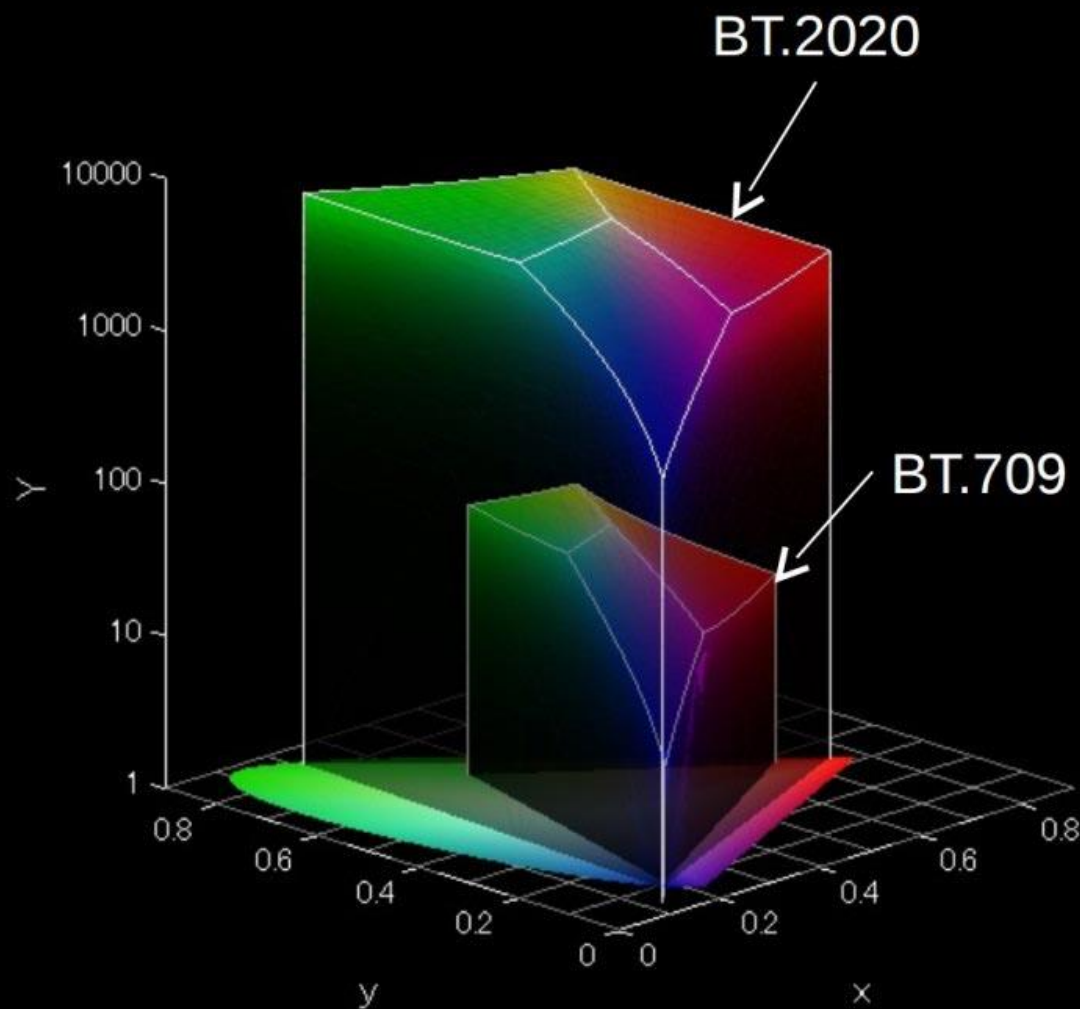
- are much higher peak brightness levels
- have much higher dynamic range (from very deep blacks to very bright highlights)

... so need to stretch or modify gamma to allow a little more brightness  
and dynamic range out the current system

... and adapt to scenes



# Colour in BT.709 vs BT.2020



Colour/luminance volume:

BT.2020 (10,000 nits for new  
LCDs and OLED displays)

vs

BT.709 (100 nits); Yxy  
(Sony)

BT.2020 can reproduce a  
much wider range of colours,  
including more saturated  
greens, cyans, blues, and  
reds.

**Nit (unit used for brightness):** Unit of  
luminance, the intensity of visible light.  
To describe the brightness of TV displays.  
One nit = one candela/m<sup>2</sup>

The height of the 3D volume  
indicates the range of  
brightness each colour can  
achieve.

# The Need for HDR Transfer Functions – HLG and PQ

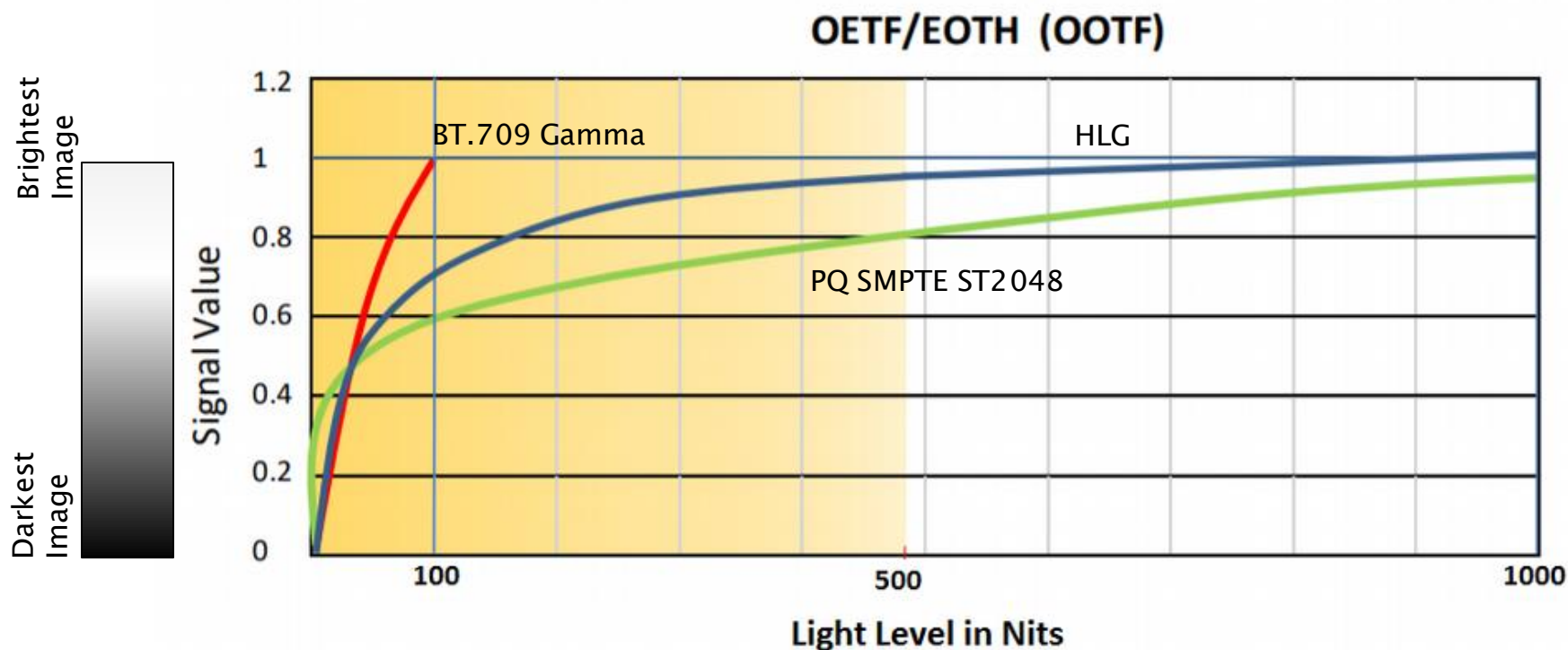
## Traditional display devices

- used BT.709 Gamma transfer function (around 2.2–2.4) for encoding and decoding brightness.
- brightness range of up to 100 Nits (cd/m<sup>2</sup>).

## For HDR

- **Perceptual Quantizer (PQ)** is a critical transfer function for HDR video. It provides a standardized, perceptually optimized method for **mapping a very wide range of absolute light levels into digital code values**, enabling the accurate representation of both deep shadows and bright highlights that characterize HDR content
- **Hybrid Log–Gamma (HLG)** – is a relative HDR transfer function that combines a **gamma curve for lower brightness levels** and **a logarithmic curve for higher brightness levels**.
- This hybrid design makes it backward compatible with SDR displays, allowing a single HLG signal to be viewed acceptably on both SDR and HDR TVs.
- This characteristic makes HLG particularly well–suited for broadcast applications, offering a simpler pathway to delivering HDR content.

# HDR – Signal To Light Mapping

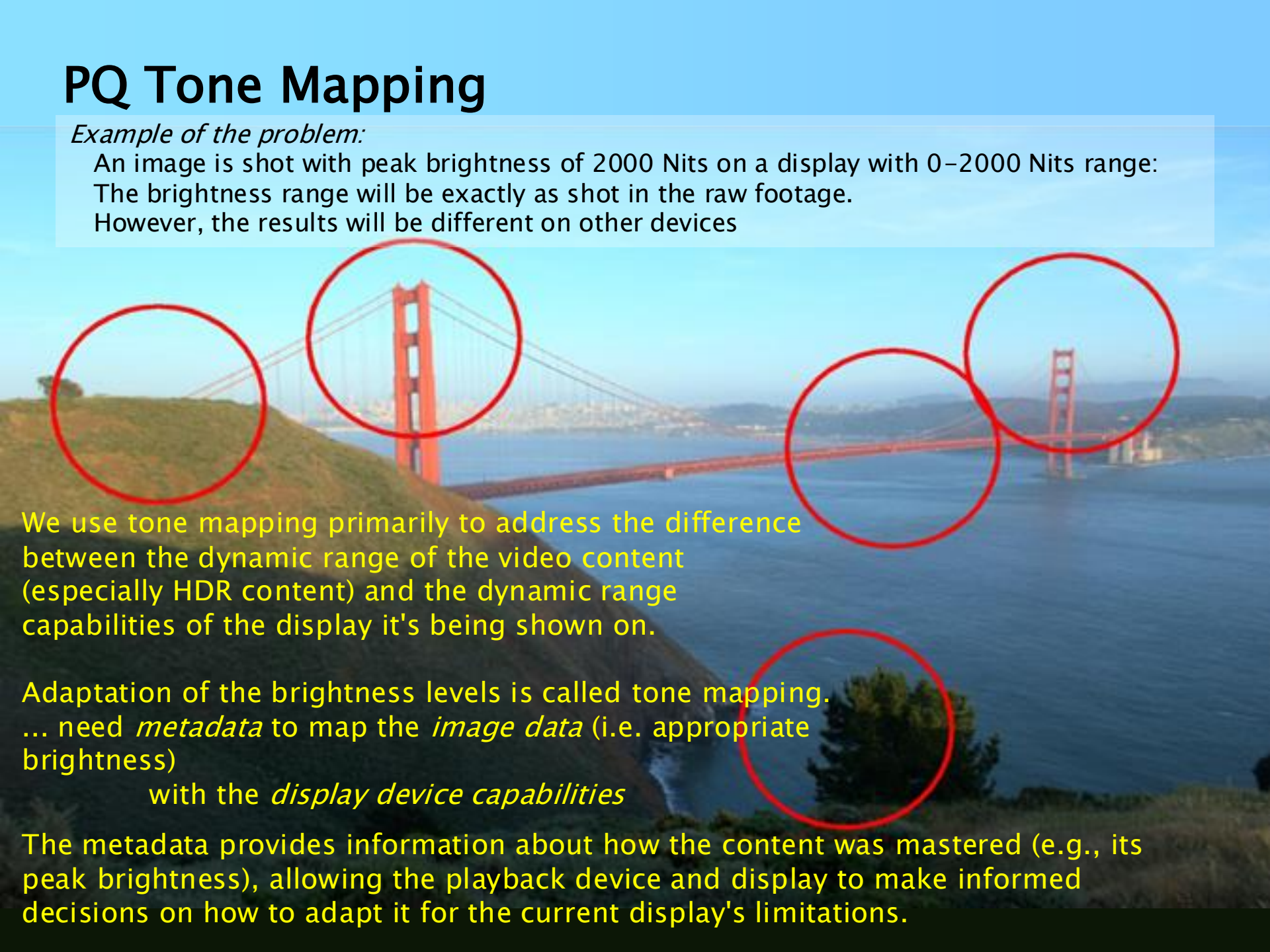


- Light level range in darker region is represented by a larger signal value range compared to the brighter regions: so more granular representation in darker regions.
- Since our eyes are more sensitive to changes in the dark, these systems try to allocate more "digital steps" to represent those dark areas accurately.
- HDR systems like PQ need to handle a much wider range of brightness, so their signal mapping needs to be different from the older systems.
- HLG tries to find a middle ground.

# PQ Tone Mapping

*Example of the problem:*

An image is shot with peak brightness of 2000 Nits on a display with 0–2000 Nits range:  
The brightness range will be exactly as shot in the raw footage.  
However, the results will be different on other devices



We use tone mapping primarily to address the difference between the dynamic range of the video content (especially HDR content) and the dynamic range capabilities of the display it's being shown on.

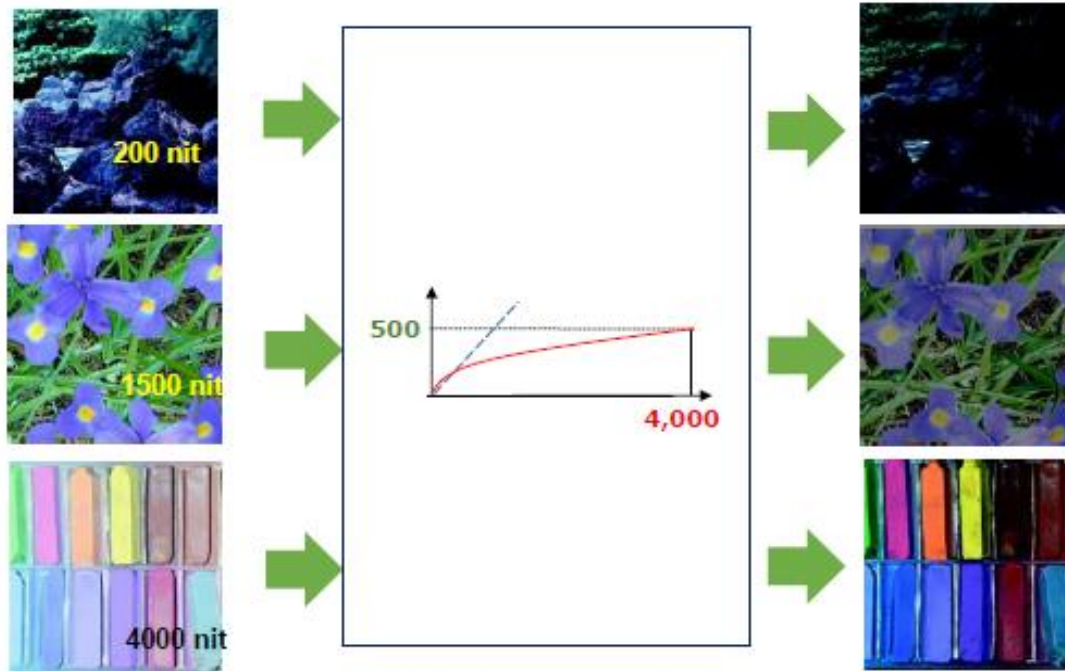
Adaptation of the brightness levels is called tone mapping.  
... need *metadata* to map the *image data* (i.e. appropriate brightness)  
with the *display device capabilities*

The metadata provides information about how the content was mastered (e.g., its peak brightness), allowing the playback device and display to make informed decisions on how to adapt it for the current display's limitations.

# SMPTE ST 2094 and Dynamic Metadata

## Static Tone Mapping – HDR10

Optimized **only** for the **brightest scene** in the contents



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Illustrations courtesy of  
YT Kim, Samsung

- **Static tone mapping**, as used in HDR10, applies a single tone mapping curve to the entire video content, optimized for the brightest scene.
- This means that darker scenes might lose detail and appear too dark, while very bright scenes might have their highlights clipped or compressed in a suboptimal way.



# SMPTE ST 2094 and Dynamic Metadata

Now, instead of a single static curve, there are three different tone mapping curves, one for each scene.

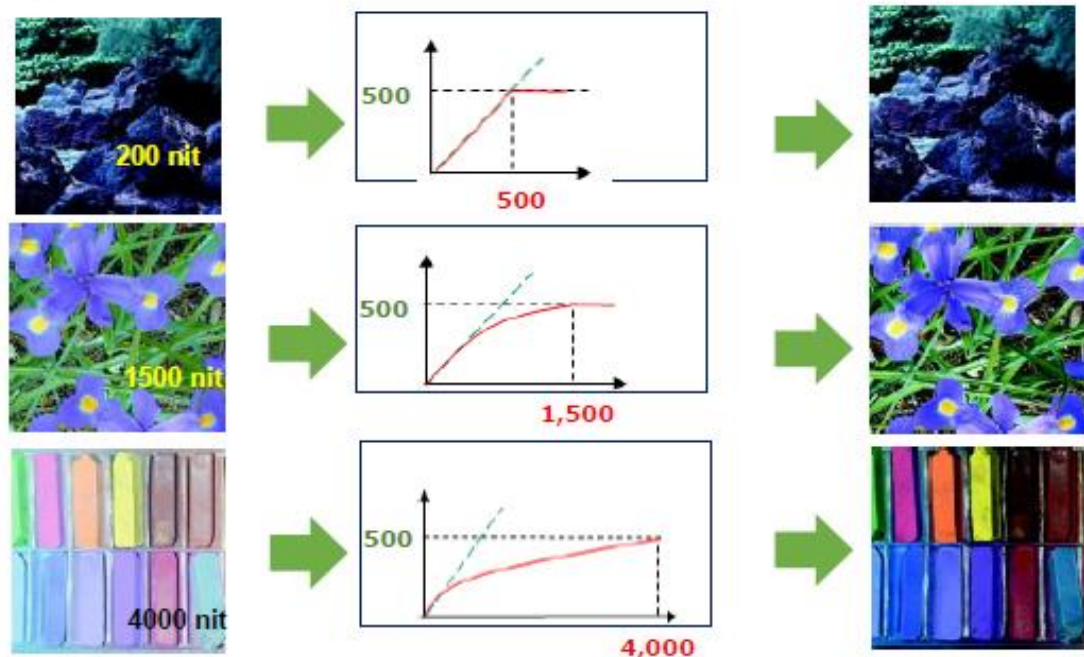
## Dynamic Tone Mapping – DMCVT

Optimizes each scene

The curve likely has a steeper initial rise to bring out detail in the darker areas, with its output range optimized for lower nit values

The curve is likely optimized for a mid-range brightness, making good use of the available signal range for this content.

The curve extends to handle the higher peak brightness, compressing the highlights appropriately for the target display



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- **Dynamic tone mapping**, enabled by dynamic metadata, allows the display to adapt its tone mapping strategy on a scene-by-scene (or even frame-by-frame) basis.
- This ensures that each part of the content is displayed with optimal brightness and contrast, preserving detail in both the shadows and highlights,

# PQ Tone Mapping

The display needs information about how the HDR content was created (e.g., its peak brightness, colour space) to perform accurate tone mapping.

Then the display can decide the tone mapping parameters  
... the video is transformed to lie optimally within the display range of the display device



The full brightness range of the HDR content is compressed into a much smaller range, likely resulting in loss of highlight detail and a less impactful image.

2000 nit content is intelligently adapted to fit within 1000 nit range, preserving more of the original dynamic range than the simple BT.709 transform.

HDR content can be displayed with its full dynamic range without significant tone mapping.

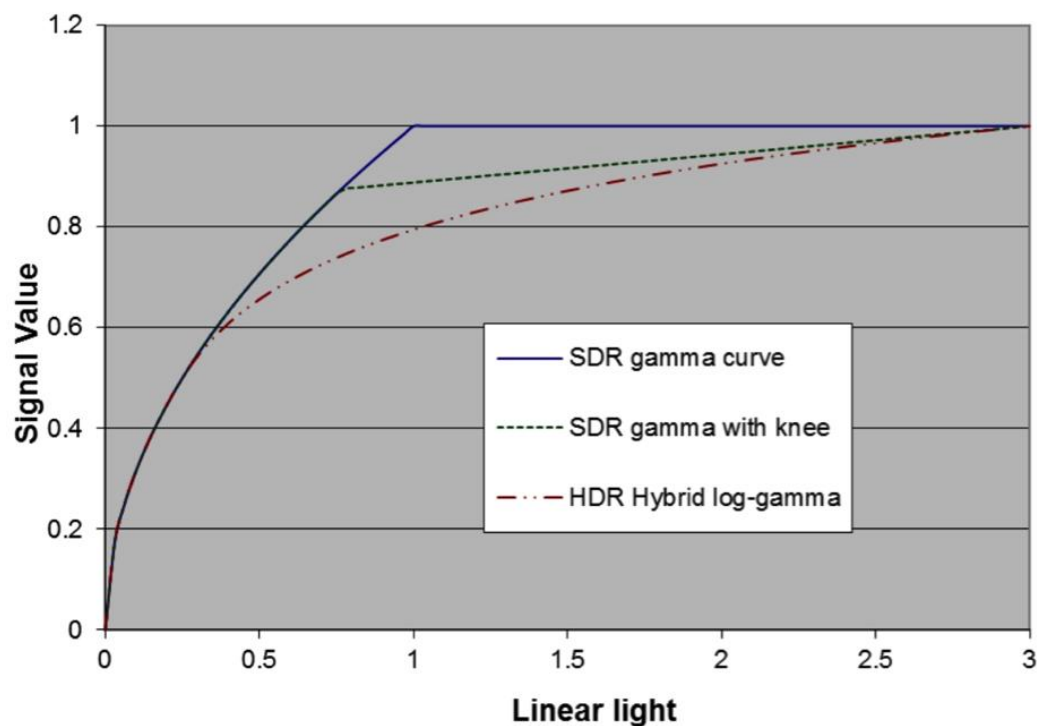
The Perceptual Quantizer (PQ) as a standard is published by SMPTE as ST2084

# Hybrid Log Gamma

A single HLG signal can be interpreted as a standard dynamic range (SDR) signal by older TVs and as a high dynamic range (HDR) signal by newer, compatible displays.

- uses a **logarithmic curve** in the highlights (since the quantisation noise is almost constant)
- and a **gamma curve** in the dark areas since the visibility threshold for quantisation is higher
- gamma curve in HLG (for a large part of the signal range)  
= backward compatibility with legacy displays

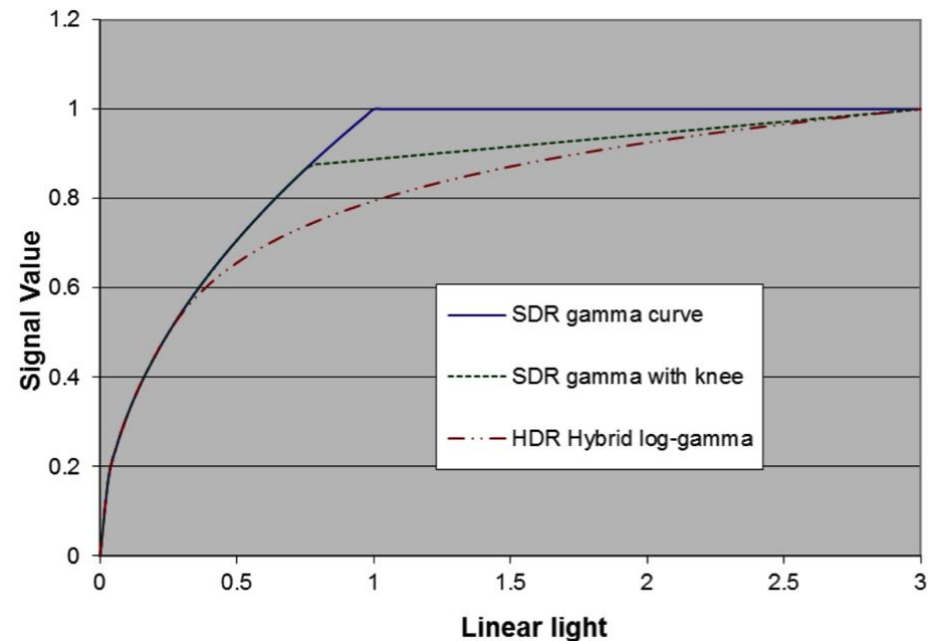
HLG refers to the fact that the OETF is a hybrid that applies a conventional gamma curve for low-light signals and a logarithmic curve for the high tones.





# Hybrid Log Gamma

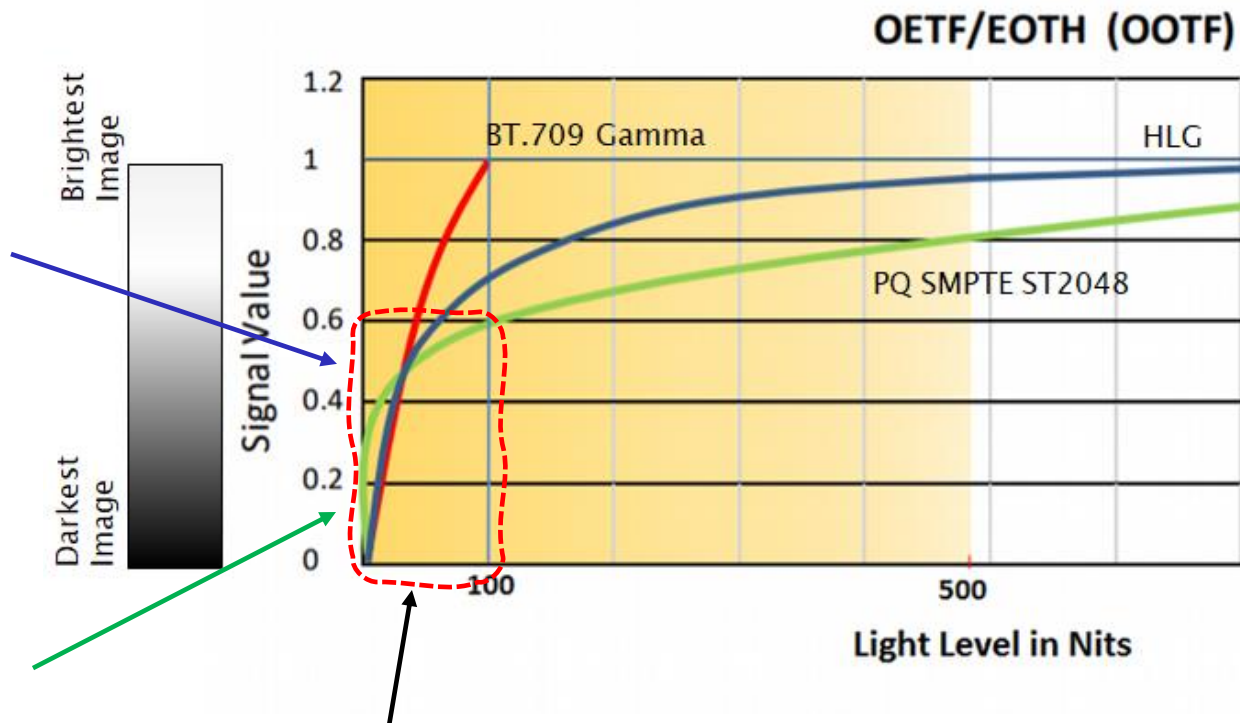
- Transfer characteristics specified by HLG can be implemented directly in the camera
- Signal produced by the camera is independent of the display
- No additional processing, no metadata
  - Unlike PQ, which often relies on metadata to guide tone mapping, HLG is designed to be interpreted reasonably well by both SDR and HDR displays without the need for extra dynamic metadata.
  - This makes it simpler for broadcasting and live TV where sending extra data might be difficult.



# HDR – Signal To Light Mapping

For HLG – more than half of signal values are represented for light level between 0–60 Nits remaining signal values are represented in 60–1000 Nits range

PQ displays are similar – approx. half of the signal values are for light level between 0–40 Nits the remaining half of signal values are represented in 40–1000 Nits range.



- HLG is similar to BT.709 in lower brightness regions – better compatibility with the existing SDR displays
- PQ is different from BT.709
  - on a standard SDR display, darker regions represented by PQ will invariably become brighter
  - ... reducing the contrast levels of the image ... washed out image

*PQ requires tone mapping by display devices*



B B C HD

1080 lines